

MAT238 Calculus III Equation Sheet:

Max and Min Info:

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} \begin{array}{l} \text{localmin : } D > 0, f_{xx}(a, b) > 0 \\ \text{localmax : } D > 0, f_{xx}(a, b) < 0 \\ \text{saddlepoint : } D < 0 \end{array}$$

Equation of a plane:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$\text{Gradient: } \nabla f = \langle f_x, f_y, f_z \rangle$$

Linearization:

$$L(x, y) = z_0 + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$$\text{Arc Length } L = \int_a^b \|\mathbf{r}'(t)\| dt$$

$$\text{Differential: } dz = f_x(x, y)dx + f_y(x, y)dy$$

$$\text{Directional Derivative: } D_{\mathbf{u}}f = \nabla f(x, y) \cdot \mathbf{u}$$

$$\text{Lagrange Multiplier: } \nabla f(x, y) = \lambda \nabla g(x, y)$$

Polar Coordinates:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

$$\text{Jac: } r$$

Cylindrical Coordinates:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$x^2 + y^2 = r^2$$

$$\text{Jac: } r$$

Spherical Coordinates:

$$x = \rho \sin \varphi \cos \theta$$

$$y = \rho \sin \varphi \sin \theta$$

$$z = \rho \cos \varphi$$

$$x^2 + y^2 + z^2 = \rho^2$$

$$\text{Jac: } \rho^2 \sin \varphi$$

$$A = \iint_R dA \quad V = \iint_R f(x, y) dA \quad V = \iiint_E dV \quad \text{Avg Value} = \frac{1}{A(R)} \iint_R f(x, y) dA$$

$$\text{Mass: } m = \iiint_E \rho(x, y, z) dV; \text{ Surface Area: } SA = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| dA = \iint_D \sqrt{(f_x)^2 + (f_y)^2 + 1} dA$$

Jacobian:

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \iint_R f(x, y) dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

Change of Variables:

Vector Field: $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$

Conservative $\mathbf{F}$: $\nabla f = \mathbf{F}$

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$$

$$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$$

Line integrals:

$$\int_C f(x, y, z) ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (\mathbf{F} \cdot \mathbf{T}) ds = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_C P dx + Q dy + R dz$$

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

Surface Integrals:

$$\iint_S f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| dA = \iint_D f(x, y, g(x, y)) \sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2} dA$$

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S (\mathbf{F} \cdot \mathbf{N}) dS = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) dA = \iint_D \left(-P \frac{\partial z}{\partial x} - Q \frac{\partial z}{\partial y} + R \right) dA$$

$$\text{Green's Theorem: } \oint_{\partial D} P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA; \quad \text{flux} = \oint_{\partial D} \mathbf{F} \cdot \mathbf{N} ds = \iint_D \text{div } \mathbf{F} dA$$

$$\text{Stokes Theorem: } \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}, \quad \text{Divergence Theorem: } \iint_{\partial E} \mathbf{F} \cdot d\mathbf{S} = \iiint_E \text{div } \mathbf{F} dV$$